# NutriTrack AI — Comprehensive System & Architectural Context

Intelligent Personal Nutrition, Computer Vision & Smart Festival Balance Platform

| | |
|---|---|
| **Production URL:** https://nutritrack-app-beryl-five.vercel.app | **Version:** 2.5 (Production Release) |
| **GitHub Repository:** github.com/pranaykumbhar31-maker/NutriTrackk-AI | **Stack:** Next.js 16 (Turbopack), TypeScript, Tailwind |
| **Author / Maintainer:** Pranay & NutriTrack AI Team | **Last Updated:** October 2026 |

## 1. Executive Summary & Core Philosophy

**NutriTrack AI** is an intelligent, clinically calibrated personal nutrition and lifestyle management web application designed to eliminate the restrictive guilt, repetitive inaccuracies, and social friction found in conventional calorie-counting tools. Centered around the principle of *'AI Nutrition, Made Human'*, NutriTrack AI bridges everyday nutrition goals with real-world social occasions, holidays, weddings, and cultural celebrations.

### The Fundamental Flaws of Traditional Trackers:

- **Rigid Lab Mindset:** Most trackers assume human life occurs in a sterile vacuum. Eating a festive treat or going to a wedding triggers punitive calorie warnings, red charts, and feelings of failure.
- **Random/Fluctuating Numbers:** AI vision tools often hallucinate erratic macro values for the same food (e.g., claiming a single apple has 30g of protein or 600 kcal).
- **Disconnection from Daily Routines:** Users are told their daily targets, but are left stranded on *how* to structure their actual day — when to eat, what portions to cook, and what workout split fits their exact goal.

### NutriTrack AI's Scientific Solution:

1. **Strict Atwater Energy Laws:** Eliminates all random hallucinations by enforcing the fundamental nutritional equation: Calories = (Protein × 4) + (Carbs × 4) + (Fats × 9), cross-verified against verified USDA and IFCT databases.
2. **3-Phase Smart Festival Intelligence:** Connects nutrition with festivals (Navratri, Diwali, Weekend Celebrations) through pre-event blood-sugar preparation, during-event mindful enjoyment with zero guilt, and post-event scientific electrolyte restoration.
3. **Automated Goal & Routine Chart Generator:** Asks for user biometrics (age, weight, height, goal, pace, workout setting) and produces a complete, timed daily meal routine (07:00 AM to 09:45 PM) and a 7-day structured exercise routine.

# 2. System Architecture & Tech Stack

| Component Layer | Technology / Implementation Details |
|---|---|
| **Core Framework** | Next.js 16.3.3 (Turbopack engine), React 19, TypeScript 5.8 with strict typing. |
| **Styling & Design System** | Tailwind CSS v4 with custom Emerald/Teal medical-wellness theme, Lucide React icons, and canvas-confetti. |
| **Computer Vision Engine** | Dual-mode architecture: Node.js sharp spectral analysis + Client-side HTML5 Canvas pixel sampling + Groq/Gemini Vision API fallbacks. |
| **Persistence Architecture** | Local-First Storage with browser event dispatchers (nutritrack_meals_updated, nutritrack_profile_updated) synced with server DB API routes (/api/user/sync). |
| **Biometric Algorithms** | Mifflin-St Jeor BMR Equation, TDEE activity multipliers (1.2 to 1.725), Quetelet BMI Index, and Atwater Specific Energy Factor System. |
| **Hosting & CI/CD** | Vercel Global Edge Network (East US - iad1), Turbopack optimized builds with 18 static/dynamic routes. |

# 3. Detailed Core Feature Breakdown

## 3.1 Trained AI Food Scanner & Computer Vision Engine

The NutriTrack AI Food Scanner allows users to snap live camera photos, upload meal images, or select realistic pre-curated meal plates. Unlike standard vision wrappers that blindly accept any uploaded photo, NutriTrack AI features a dedicated **Dual-Gate Validation System**:

- **Strict Non-Food Rejection Guard:** Analyzes image entropy, color variance, and RGB distribution. If the image depicts text documents, screenshots, receipts, mobile phones, laptops, selfie faces, clothing, tools, or vehicles, scanning is immediately rejected with: *'Non-food item detected. NutriTrack AI only scans edible food items, meals, and beverages.'*
- **Single-Food Item Acceptance:** Simple, single-ingredient foods (e.g., an apple, banana, boiled egg, bowl of steamed white rice, bread slice) are cleanly recognized without false negative errors.
- **Multi-Item Plate Decomposition:** Complex dishes are segmented into constituent foods (e.g. Basmati Rice 150g + Dal Tadka 180g + Palak Paneer 80g).
- **Sticky Query Elimination:** Fresh file uploads and camera snapshots automatically clear previous query states, preventing old search terms from hijacking new photo scans.

## 3.2 Personalized Goal & Daily Routine Chart Generator (/routine)

A marquee addition to NutriTrack AI, the Goal & Routine Chart Generator transforms abstract health targets into an actionable, daily operational schedule:

- **Personalized Intake Parameters:** Captures Name, Age, Gender, Weight, Height, Target Weight, Primary Goal (Fat Loss, Muscle Hypertrophy, Maintenance, Longevity), Weekly Pace, Activity Level, Dietary Preference (Veg, Non-Veg, Vegan, Eggetarian, Keto), Allergies, and Workout Location (Gym, Home Dumbbells, Bodyweight).
- **Biometric Target Calculation:** Derives exact BMR, TDEE, calibrated Daily Calorie target, Protein (g), Carbohydrates (g), Healthy Fats (g), Fiber (g), Hydration allowance (Liters and 250ml cups), and projected weeks to completion.
- **6-7 Timed Daily Meal Protocol:** Features Early Morning Kickstart (07:00 AM), Power Breakfast (08:30 AM), Mid-Morning Snack (11:30 AM), Balanced Lunch (01:30 PM), Evening Fuel (05:00 PM), Restorative Dinner (08:00 PM), and Bedtime Elixir (09:45 PM), complete with portion weights, macro breakdowns, clinical benefits, and smart culinary swaps.
- **7-Day Structured Workout Routine:** Distinct splits (Push, Pull, Legs, Active Recovery, Full Body Conditioning, Cardio, Rest) with warm-up sets, 4-5 targeted exercises with Sets, Reps, Rest intervals, target muscles, form tips, and cool-down protocols.

• **Interactive Daily Adherence Checklist:** Live check-off system for hydration milestones, workouts, and meal logging with progress tracking and confetti celebration.

• **One-Click Dashboard Sync:** The *'Apply to Dashboard'* button instantly writes calculated targets to user storage, re-calibrating the dashboard energy gauge and macro split bars.

### 3.3 Smart Nutrition Calendar & Festival Mode (/calendar)

Cultural festivals, family dining, and celebrations are an essential part of life. Traditional dieting approaches lead to binge-and-starve cycles. NutriTrack AI resolves this through its **3-Phase Festive Lifecycle Architecture**:

| Phase | Strategic Objective | Nutritional Action Protocol |
|---|---|---|
| **Phase 1: Pre-Event Preparation** | Blood Sugar Stabilization & Satiety Anchoring | Higher morning protein (Greek yogurt, eggs, paneer) and dietary fiber to blunt insulin spikes. Pre-event hydration of 1.5L - 2.0L water. |
| **Phase 2: During-Event Mindful Enjoyment** | Guilt-Free Presence & Savoring Favorites | 1-tap logging of festive foods (steamed modaks, ladoos, biryani). Zero punitive alarms or shame-based notifications. |
| **Phase 3: Post-Event Scientific Balance** | Electrolyte Balance & Normal Routine Resumption | No extreme fasting or compensatory starvation. Potassium-rich meals (spinach, avocado, lentils) and gentle 20-30 min walking to naturally balance temporary sodium retention. |

### 3.4 Interactive Daily Dashboard (/dashboard)

- **Energy Left Donut Gauge:** Displays real-time consumed vs remaining calories with smooth radial SVG progress and color transitions.
- **Macronutrient Distribution Split:** Visual progress bars for Protein, Carbohydrates, and Fats with remaining gram targets.
- **Interactive Water Logging:** Track glasses and liters against daily targets with quick increment/decrement buttons.
- **Categorized Meal Cards:** Breakdown by Breakfast, Lunch, Dinner, and Snacks with photo thumbnails, macro tags, portion adjustments, and deletion.
- **Dynamic Quetelet BMI Status:** Computes live BMI and healthy target weight bracket based on current user height and weight.

### 3.5 Conversational Assistant & Smart Reminder Drawer

- **NutriTrack AI Coach:** Floating assistant bubble and popover drawer providing empathetic advice for dining out, overeating recovery, and high-protein recipe recommendations.
- **Smart Reminders:** Sliding drawer with toggleable notifications for breakfast, lunch, dinner, water milestones, and festival preparation alerts.

### 3.6 Multi-Profile Architecture & Hybrid Backend Persistence

- **Multi-Account Management:** Allows multiple family members or demo profiles to switch seamlessly from the navigation bar dropdown.
- **Hybrid Data Synchronization:** LocalStorage provides instantaneous offline-first responsiveness; background API calls to /api/user/sync sync account state with server persistence.

# 4. Project Architecture & File Directory Map

| Path | Type | Core Responsibility |
|---|---|---|
| `src/app/page.tsx` | Page (Client) | Root Home page with Hero section, Goal & Routine spotlight, Festival feature cards, and navigation links. |
| `src/app/routine/page.tsx` | Page (Client) | Interactive Goal & Routine Generator page. Displays 6-meal daily schedule, 7-day workout routine, and adherence checklist. |
| `src/app/scanner/page.tsx` | Page (Client) | AI Food Scanner page with live camera, photo upload, preset dishes, and non-food validation warnings. |
| `src/app/dashboard/page.tsx` | Page (Client) | Daily nutrition tracking center: Energy Left gauge, meal slots, water intake, and BMI tracking. |
| `src/app/calendar/page.tsx` | Page (Client) | Calendar & Festival intelligence center with 3-phase pre/during/post-event guidance. |
| `src/app/recommendations/page.tsx` | Page (Client) | Target-calibrated meal suggestions and workout recommendations. |
| `src/app/analytics/page.tsx` | Page (Client) | Calorie streaks, weight trajectory charts, and festival recovery analysis. |
| `src/app/login/page.tsx` | Page (Client) | User sign in, registration wizard, and demo account switching. |
| `src/app/profile/page.tsx` | Page (Client) | Biometric stats, allergy management, and calorie target calibration. |
| `src/lib/routineEngine.ts` | Library | Algorithmic engine generating timed diet plans, 7-day workout splits, and daily habit checklists. |
| `src/lib/foodVisionEngine.ts` | Library | Spectral computer vision classifier, non-food rejection guard, and Atwater nutrient calibration. |
| `src/lib/festivalEngine.ts` | Library | Festival lifecycle algorithms, pre/post event logic, and Event Impact Reports. |
| `src/lib/nutritionEngine.ts` | Library | Mifflin-St Jeor BMR, TDEE, BMI, and daily macro aggregation calculations. |
| `src/lib/storage.ts` | Library | Local storage state management, event dispatchers, and backend sync integration. |
| `src/lib/foodDatabase.ts` | Library | Comprehensive verified database of 150+ foods with USDA/IFCT macro profiles. |
| `src/app/api/scan-food/route.ts` | API Route | Food scanning POST endpoint: verifies image validity, performs vision analysis, and returns Atwater macros. |
| `src/app/api/assistant/route.ts` | API Route | Nutrition coaching chat endpoint with ethical guidelines and festive support. |
| `src/app/api/auth/...` | API Route | Backend user registration, login, and user list endpoints. |
| `src/app/api/user/sync/route.ts` | API Route | Endpoint for syncing user profile, meals, water, and events to server database. |

# 5. API Architecture & Request/Response Specification

| Endpoint | Method | Payload Parameters | Output Description |
|---|---|---|---|
| `/api/scan-food` | POST | imageBase64, sampleDishId, fileName, userDescription | Returns isFood: boolean, detected food items, Atwater nutrition, health score, and clinical observations. Rejects non-food with explanation. |
| `/api/assistant` | POST | message, history, userProfile, todayConsumed, upcomingEventName | Returns empathetic, scientifically grounded conversational advice tailored to the user's remaining calories and festive context. |
| `/api/auth/register` | POST | name, email, password, profile | Creates a new user account and profile record in server persistence. |
| `/api/auth/login` | POST | email, password | Authenticates user credentials and returns full account state and profile. |
| `/api/user/sync` | POST | userId, payload: { profile, meals, water, events } | Synchronizes client-side local changes to the server database for cross-device continuity. |

# 6. Installation, Local Execution & Deployment

**Prerequisites:** Node.js v18.0.0 or higher (v24 LTS recommended) and npm v9+ installed.

### 1. Clone & Install Dependencies:

```
git clone https://github.com/pranaykumbhar31-maker/NutriTrackk-AI.git
cd NutriTrackk-AI
npm install
```

### 2. Run Development Server:

npm run dev

Access the application at http://localhost:3000 with instant Turbopack hot reloading.

### 3. Production Build & Verification:

npm run build

npm start

Verifies all 18 routes, strict TypeScript types, and server-side compilation.

### 4. Vercel Production Deployment:

npx vercel --prod --yes

Deploys directly to the live production domain with edge distribution and zero downtime.

# 7. Strategic Value & Future Enhancements

NutriTrack AI successfully establishes a new paradigm in digital nutrition. By combining **Atwater scientific integrity**, **intelligent non-food computer vision validation**, **empathetic festival lifecycle balance**, and **automated goal-driven routine charts**, it provides users with an actionable roadmap to sustainable health without neurotic restriction.

- **Upcoming Enhancements:** Apple Health & Google Health Connect wearable sync, multi-lingual Indian regional language support, and barcode scanning for packaged grocery items.